

Chapter 9 — State, Time, Events, Messaging, Coordination, and Persistence

Purpose and scope

Defines authoritative state, projections, commands, queries, events, ordering, time, idempotency, messaging, and durable persistence.

Source authority

Book I: Book I evidence, continuity, reversibility, accountability, and failure containment

Book II: Book II Chapters 13, 22, 23, 24, 25, 27, and 28

Book III: Book III Chapters 03, 04, 06, and 07

Normative semantic rules

- Commands **MUST** request possible change; Events **MUST** state completed facts; Queries **MUST NOT** intentionally mutate authoritative state.
- Each authoritative state domain **MUST** identify one mutation owner; projections, caches, and replicas **MUST** remain explicitly derived.
- Ordering claims **MUST** state scope and mechanism; Wall-Clock Time **MUST NOT** be used as sole proof of distributed causality.
- Idempotency claims **MUST** state operation, key scope, retention horizon, result semantics, and external-effect limitations.

Canonical terms

HAL-TERM-0101 — Command

A request addressed to an authoritative owner to evaluate and, if allowed, perform a state-changing operation.

Distinction: A Command is not evidence that the operation was authorized, committed, or completed.

Type/category/status: Intent message / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0101` when it uses ****Command**** with this exact governed meaning: A request addressed to an authoritative owner to evaluate and, if allowed, perform a state-changing operation.

Counterexample: A dependent artifact uses ****Command**** in a way that violates its required distinction: A Command is not evidence that the operation was authorized, committed, or completed.

Relationships: HAL-REL-0037, HAL-REL-0040

Lifecycle: None registered

Constraint: A Command is not evidence that the operation was authorized, committed, or completed.

HAL-TERM-0102 — Query

A request for information that must not intentionally mutate authoritative state or external reality.

Distinction: Telemetry side effects must remain non-authoritative and must not turn a Query into a hidden Command.

Type/category/status: Read request / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0102` when it uses ****Query**** with this exact governed meaning: A request for information that must not intentionally mutate authoritative state or external reality.

Counterexample: A dependent artifact uses ****Query**** in a way that violates its required distinction: Telemetry side effects must remain non-authoritative and must not turn a Query into a hidden Command.

Relationships: None registered

Lifecycle: None registered

Constraint: Telemetry side effects must remain non-authoritative and must not turn a Query into a hidden Command.

HAL-TERM-0103 — Event

An immutable message stating that a defined fact or state transition has completed in its owning domain.

Distinction: An Event is not a request, intention, or mutable current-state record.

Type/category/status: Completed-fact record / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0103` when it uses ****Event**** with this exact governed meaning: An immutable message stating that a defined fact or state transition has completed in its owning domain.

Counterexample: A dependent artifact uses ****Event**** in a way that violates its required distinction: An Event is not a request, intention, or mutable current-state record.

Relationships: HAL-REL-0037, HAL-REL-0039

Lifecycle: None registered

Constraint: An Event is not a request, intention, or mutable current-state record.

HAL-TERM-0104 — Event Journal

The ordered durable record of Events for an owning domain, with identity, sequence, provenance, and integrity controls.

Distinction: Global total order must not be inferred when only per-stream ordering exists.

Type/category/status: Durable event store / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0104` when it uses ****Event Journal**** with this exact governed meaning: The ordered durable record of Events for an owning domain, with identity, sequence, provenance, and integrity controls.

Counterexample: A dependent artifact uses ****Event Journal**** in a way that violates its required distinction: Global total order must not be inferred when only per-stream ordering exists.

Relationships: HAL-REL-0038

Lifecycle: None registered

Constraint: Global total order must not be inferred when only per-stream ordering exists.

HAL-TERM-0105 — Message Envelope

The governed wrapper carrying message identity, schema version, source, destination, time, correlation, causation, authority context, classification, and integrity metadata.

Distinction: The envelope does not change the semantic type of its payload.

Type/category/status: Transport record / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0105` when it uses ****Message Envelope**** with this exact governed meaning: The governed wrapper carrying message identity, schema version, source, destination, time, correlation, causation, authority context, classification, and integrity metadata.

Counterexample: A dependent artifact uses ****Message Envelope**** in a way that violates its required distinction: The envelope does not change the semantic type of its payload.

Relationships: HAL-REL-0040

Lifecycle: None registered

Constraint: The envelope does not change the semantic type of its payload.

HAL-TERM-0106 — Idempotency Key

A scoped stable value allowing a receiver to recognize equivalent retry Attempts for one declared operation and result horizon.

Distinction: It does not make non-idempotent external effects reversible or globally exactly-once.

Type/category/status: Deduplication value / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0106` when it uses ****Idempotency Key**** with this exact governed meaning: A scoped stable value allowing a receiver to recognize equivalent retry Attempts for one declared operation and result horizon.

Counterexample: A dependent artifact uses ****Idempotency Key**** in a way that violates its required distinction: It does not make non-idempotent external effects reversible or globally exactly-once.

Relationships: None registered

Lifecycle: None registered

Constraint: It does not make non-idempotent external effects reversible or globally exactly-once.

HAL-TERM-0107 — Correlation Identifier

A value linking related work, messages, evidence, and telemetry across a bounded flow.

Distinction: Correlation does not prove causation.

Type/category/status: Trace reference / Distributed systems / Approved

Aliases: correlation ID

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0107` when it uses ****Correlation Identifier**** with this exact governed meaning: A value linking related work, messages, evidence, and telemetry across a bounded flow.

Counterexample: A dependent artifact uses ****Correlation Identifier**** in a way that violates its required distinction:

Correlation does not prove causation.

Relationships: None registered

Lifecycle: None registered

Constraint: Correlation does not prove causation.

HAL-TERM-0108 — Causation Identifier

A value identifying the immediate initiating message, decision, or event for a derived operation.

Distinction: It records declared lineage and must not be used as sole proof of real-world causality.

Type/category/status: Causal reference / Distributed systems / Approved

Aliases: causation ID

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40

Book II: Book II Chapters 22 and 23

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0108` when it uses ****Causation Identifier**** with this exact governed meaning: A value identifying the immediate initiating message, decision, or event for a derived operation.

Counterexample: A dependent artifact uses ****Causation Identifier**** in a way that violates its required distinction: It records declared lineage and must not be used as sole proof of real-world causality.

Relationships: None registered

Lifecycle: None registered

Constraint: It records declared lineage and must not be used as sole proof of real-world causality.

HAL-TERM-0109 — Authoritative State

State whose mutation and truth are governed by the designated owning domain.

Distinction: A cache, replica, projection, search index, or local aggregate is not authoritative unless explicitly designated.

Type/category/status: Owned state / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40

Book II: Book II Chapters 23 and 24

Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0109` when it uses ****Authoritative State**** with this exact governed meaning: State whose mutation and truth are governed by the designated owning domain.

Counterexample: A dependent artifact uses ****Authoritative State**** in a way that violates its required distinction: A cache,

replica, projection, search index, or local aggregate is not authoritative unless explicitly designated.

Relationships: None registered

Lifecycle: None registered

Constraint: A cache, replica, projection, search index, or local aggregate is not authoritative unless explicitly designated.

HAL-TERM-0110 — Projection

A read-optimized representation derived from authoritative records or Events for a declared purpose.

Distinction: A Projection may be stale and must not silently accept authoritative mutations.

Type/category/status: Derived state / Distributed systems / Approved

Aliases: read model

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40

Book II: Book II Chapters 23 and 24

Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0110` when it uses **Projection** with this exact governed meaning: A read-optimized representation derived from authoritative records or Events for a declared purpose.

Counterexample: A dependent artifact uses **Projection** in a way that violates its required distinction: A Projection may be stale and must not silently accept authoritative mutations.

Relationships: HAL-REL-0038

Lifecycle: None registered

Constraint: A Projection may be stale and must not silently accept authoritative mutations.

HAL-TERM-0111 — Cache

A replaceable performance-oriented copy whose loss does not destroy authoritative truth.

Distinction: A Cache must not become the only copy of required state or evidence.

Type/category/status: Disposable derived state / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40

Book II: Book II Chapters 23 and 24

Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0111` when it uses **Cache** with this exact governed meaning: A replaceable performance-oriented copy whose loss does not destroy authoritative truth.

Counterexample: A dependent artifact uses **Cache** in a way that violates its required distinction: A Cache must not become the only copy of required state or evidence.

Relationships: None registered

Lifecycle: None registered

Constraint: A Cache must not become the only copy of required state or evidence.

HAL-TERM-0112 — Replica

A governed copy maintained from an authoritative source under declared consistency, lag, and failover rules.

Distinction: Replication alone does not transfer semantic ownership.

Type/category/status: Replicated state / Distributed systems / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40

Book II: Book II Chapters 23 and 24

Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0112` when it uses **Replica** with this exact governed meaning: A governed copy maintained from an authoritative source under declared consistency, lag, and failover rules.

Counterexample: A dependent artifact uses **Replica** in a way that violates its required distinction: Replication alone does not transfer semantic ownership.

Relationships: None registered

Lifecycle: None registered

Constraint: Replication alone does not transfer semantic ownership.

HAL-TERM-0113 — Transactional Outbox

A durable pattern that records an authoritative state change and its pending Event publication in one local commit boundary.

Distinction: It reduces dual-write failure but does not guarantee global exactly-once processing.

Type/category/status: Publication pattern / Distributed systems / Approved

Aliases: outbox

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40

Book II: Book II Chapters 23 and 24

Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0113` when it uses ****Transactional Outbox**** with this exact governed meaning: A durable pattern that records an authoritative state change and its pending Event publication in one local commit boundary.

Counterexample: A dependent artifact uses ****Transactional Outbox**** in a way that violates its required distinction: It reduces dual-write failure but does not guarantee global exactly-once processing.

Relationships: HAL-REL-0039

Lifecycle: None registered

Constraint: It reduces dual-write failure but does not guarantee global exactly-once processing.

HAL-TERM-0114 — Logical Time

A non-wall-clock ordering value used to represent causal or domain sequence relationships.

Distinction: Logical Time must not be presented as real elapsed or calendar time.

Type/category/status: Ordering construct / Temporal / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 44

Book II: Book II Chapter 13

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0114` when it uses ****Logical Time**** with this exact governed meaning: A non-wall-clock ordering value used to represent causal or domain sequence relationships.

Counterexample: A dependent artifact uses ****Logical Time**** in a way that violates its required distinction: Logical Time must not be presented as real elapsed or calendar time.

Relationships: None registered

Lifecycle: None registered

Constraint: Logical Time must not be presented as real elapsed or calendar time.

HAL-TERM-0115 — Wall-Clock Time

A timestamp from a declared clock source, precision, and synchronization context.

Distinction: Wall-Clock Time alone must not establish distributed causal order.

Type/category/status: Temporal observation / Temporal / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 44

Book II: Book II Chapter 13

Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0115` when it uses ****Wall-Clock Time**** with this exact governed meaning: A timestamp from a declared clock source, precision, and synchronization context.

Counterexample: A dependent artifact uses ****Wall-Clock Time**** in a way that violates its required distinction: Wall-Clock Time alone must not establish distributed causal order.

Relationships: None registered

Lifecycle: None registered

Constraint: Wall-Clock Time alone must not establish distributed causal order.

Relationship and lifecycle rules

Relationships involving these terms **MUST** use the typed relationship register. Lifecycle-bearing concepts **MUST** use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. "None registered" is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.